

POLI 784 - Problem Set #6

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```
# Load packages
library("tidyverse")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.5      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library("plm")
```

Attaching package: 'plm'

The following objects are masked from 'package:dplyr':

between, lag, lead

```
library("sandwich")
library("panelView")
```

```
## See https://yiqingxu.org/packages/panelview/ for more info.
## Report bugs -> yiqingxu@stanford.edu.
```

```
library("fixest")  
library("fect")
```

```
Registered S3 method overwritten by 'GGally':  
  method from  
+ .gg      ggplot2
```

```
Attaching package: 'fect'
```

```
The following object is masked from 'package:panelView':
```

```
simdata
```

```
library("sensemakr")
```

See details in:

Carlos Cinelli and Chad Hazlett (2020). Making Sense of Sensitivity: Extending Omitted Variables

```
library("glue")
```

Question 1

In this question, we analyze data from a study by [Sances \(2016\)](#), which evaluated the effect of a town switching from an elected to an appointed tax assessor on the frequency of property tax reassessments in New York state during the 1980s through early 2000s.

Part A

```
# Load data  
data <- read.csv("data1-2.csv")  
  
# Print number of unique units  
num_units <- length(unique(data$swis_code))  
glue("Number of units: {num_units}")
```

```
Number of units: 510
```

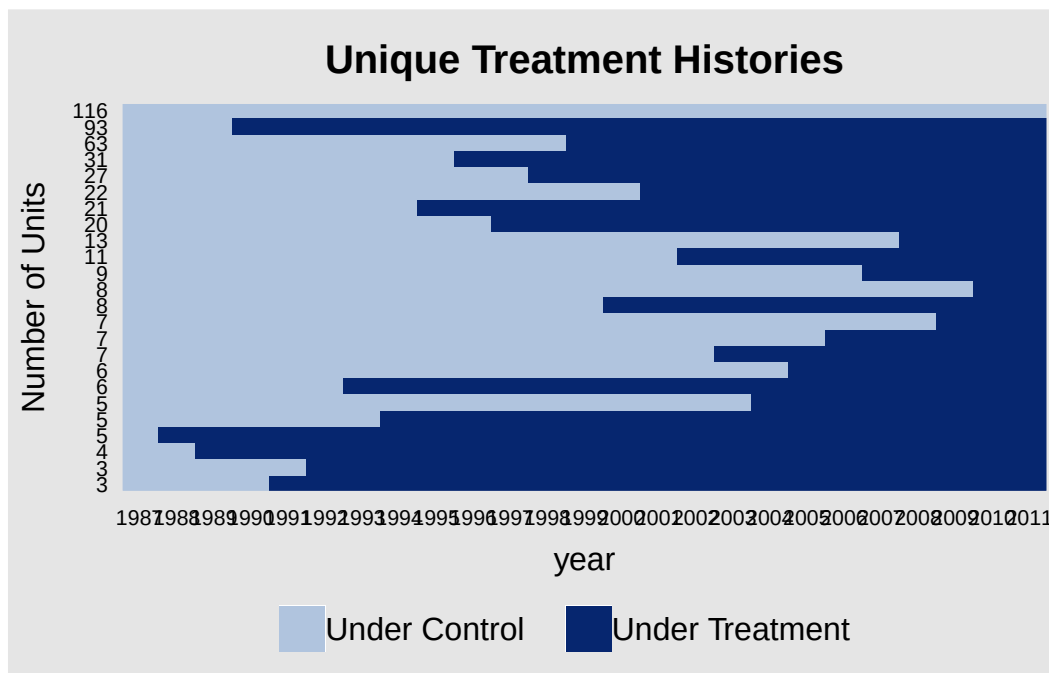
```
# Print number of unique periods
num_periods <- length(unique(data$year))
glue("Number of periods: {num_periods}")
```

Number of periods: 25

```
# Visualize treatment status using panelview
panelview(data=data,
          formula=reassessed~appointed,
          index=c("swis_code", "year"))
```

If the number of units is more than 300, we set "gridOff = TRUE".

If the number of units is more than 500, we randomly select 500 units to present.
You can set "display.all = TRUE" to show all units.



Part B

Here, we estimate the effect of switching to an appointed assessor using a classic 2x2 DiD design that compares towns that switched in 1999 to those that never switched.

```

# Filter data to include only towns that switched in 1999 (treated)
# and those that never switched (controls)
dat_DID <- data[data$cohort %in% c(1,14),]

# Get outcome variable
Y <- dat_DID$reassessed

# Create a treated/control indicator
treated <- (dat_DID$cohort == 14)
control <- !treated

# Create a pre/post treatment indicator
post_treatment <- (dat_DID$year >= 1999)
pre_treatment <- !post_treatment

# Calculate ATT using classic 2x2 DiD estimator
Ybar_treated_pre <- mean(Y[treated&pre_treatment])
Ybar_treated_post <- mean(Y[treated&post_treatment])
Ybar_control_pre <- mean(Y[control&pre_treatment])
Ybar_control_post <- mean(Y[control&post_treatment])
ATT_1B <- (Ybar_treated_post - Ybar_treated_pre) - (Ybar_control_post - Ybar_control_pre)

glue("ATT estimate produced using classic 2x2 DiD estimator: {round(ATT_1B,6)}")

```

ATT estimate produced using classic 2x2 DiD estimator: 10.680309

Part C

Here, we estimate the ATT using OLS regression of two-way transformed variables

```

# Create function to apply two-way transformation
two_way_transformation <- function(df,outcome_col,unit_col,period_col){

  # Get relevant columns
  df <- df[,c(outcome_col,unit_col,period_col)]
  colnames(df) <- c("Y","i","t")

  # Calculate unit- and period-specific means
  unit_means <- df %>% group_by(i) %>% summarise(unit_mean = mean(Y))
  period_means <- df %>% group_by(t) %>% summarise(period_mean = mean(Y))
}

```

```

# Attach to dataframe
df <- left_join(df,unit_means,by="i")
df <- left_join(df,period_means,by="t")

# Calculate grand mean
grand_mean <- mean(df$Y)

# Apply two-way transformation
Y_tw <- df$Y - df$unit_mean - df$period_mean + grand_mean

# Return result
return(Y_tw)
}

# Apply transformation to outcome and treatment columns
dat_DID$reassessed_tw <- two_way_transformation(dat_DID,"reassessed","swis_code","year")
dat_DID$appointed_tw <- two_way_transformation(dat_DID,"appointed","swis_code","year")

# Estimate ATT via OLS
# (Because we applied two-way transformation, this should be equivalent to TWFE model)
mod_1C <- lm(reassessed_tw ~ appointed_tw, data = dat_DID)
ATT_1C <- mod_1C$coefficients["appointed_tw"]
summary(mod_1C)

```

Call:

```
lm(formula = reassessed_tw ~ appointed_tw, data = dat_DID)
```

Residuals:

Min	1Q	Median	3Q	Max
-66.923	-18.289	-6.923	9.102	111.205

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.648e-17	4.645e-01	0.000	1
appointed_tw	1.068e+01	1.947e+00	5.485	4.36e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.33 on 4548 degrees of freedom

Multiple R-squared: 0.006571, Adjusted R-squared: 0.006353

F-statistic: 30.08 on 1 and 4548 DF, p-value: 4.36e-08

```
glue("ATT estimate produced using OLS regression and two-way transformation: {round(ATT_1B,6)}")
```

ATT estimate produced using OLS regression and two-way transformation: 10.680309

The point estimates of the ATT produced in Part C are equivalent to those produced in Part B. This is reassuring since the regression in Part C is equivalent to fitting a two-way fixed effects (TWFE) model, which is equivalent to DiD in settings where all treated units receive treatment at the same time. Because we specifically filtered our data to include only towns that never switch (controls) and those that switch in 1999 (treated), this criteria is satisfied and we do not need to worry about “forbidden comparisons” or “negative weights”.

Part D

Here, we use the `feols` package to fit a two-way fixed effects (TWFE) model. In the first model (shown below) we assume that errors are independently and identically distributed (i.i.d.) when computing standard errors, which is a common assumption in OLS regression.

```
## Model (1): i.i.d. errors (common assumption for OLS)
mod_1D_iid <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = dat_DID,
  vcov = "iid"
)

# Extract ATT estimate and SE
ATT_1D_iid <- mod_1D_iid$coefficients["appointed"]
SE_1D_iid <- mod_1D_iid$se["appointed"]

# print results to console
summary(mod_1D_iid)
```

```
OLS estimation, Dep. Var.: reassessed
Observations: 4,550
Fixed-effects: swis_code: 182, year: 25
Standard-errors: IID
      Estimate Std. Error t value Pr(>|t|)
appointed  10.6803    1.99263  5.3599 8.7586e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 31.3      Adj. R2: 0.274887
              Within R2: 0.006571
```

```
glue("ATT estimate produced using feols with iid errors: {round(ATT_1D_iid,6)}")
```

ATT estimate produced using feols with iid errors: 10.680309

```
glue("Standard error: {round(SE_1D_iid,6)}")
```

Standard error: 1.992632

As expected, our point estimates of the ATT are equivalent to those produced in Parts B and C. However, because the error terms of individual units are likely to be autocorrelated (e.g., towns that conduct a reassessment in year t are unlikely to conduct another reassessment in year $t + 1$) the assumption of i.i.d. errors is likely violated. To overcome this issue, we can instead compute standard errors that are clustered at the unit-level.

```
## Model (2): clustered errors (more realistic for panel data)
mod_1D_clust <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = dat_DID,
  vcov = cluster ~ swis_code
)

# Extract ATT estimate and SE
ATT_1D_clust <- mod_1D_clust$coefficients["appointed"]
SE_1D_clust <- mod_1D_clust$se["appointed"]

# print results to console
summary(mod_1D_clust)
```

```
OLS estimation, Dep. Var.: reassessed
Observations: 4,550
Fixed-effects: swis_code: 182, year: 25
Standard-errors: Clustered (swis_code)
      Estimate Std. Error t value Pr(>|t|)
appointed  10.6803    4.06961  2.6244 0.0094217 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 31.3      Adj. R2: 0.274887
      Within R2: 0.006571
```

```
glue("ATT estimate produced using feols with iid errors: {round(ATT_1D_clust,6)}")
```

ATT estimate produced using feols with iid errors: 10.680309

```
glue("Standard error: {round(SE_1D_clust,6)}")
```

Standard error: 4.069614

The above results suggest that standard OLS regression (which assumes i.i.d. errors) has a tendency to underestimate standard errors when autocorrelation is present. As such, the recommended practice is to cluster errors at the unit level when working with panel data.

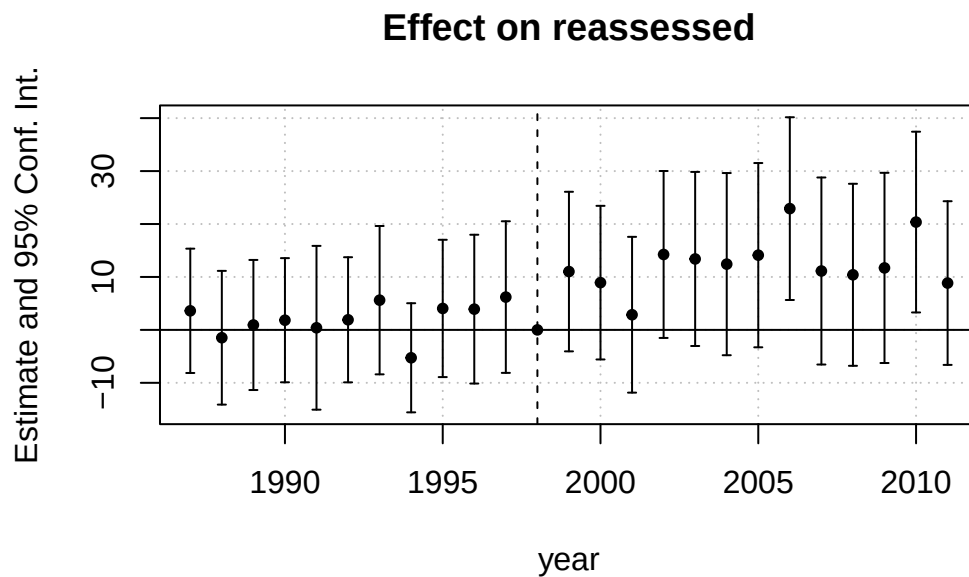
Part E

Here, we fit a leads-and-lags model to check for violations of the parallel trends assumption.

```
# Create binary variable indicating units that receive treatment
dat_DID$treated <- as.numeric(dat_DID$cohort==14)

# fit leads-and-lags model
mod_1E <- feols(
  fml = reassessed ~ i(year, treated, 1998) | swis_code + year,
  data = dat_DID,
  vcov = cluster ~ swis_code
)

# plot results
iplot(mod_1E)
```

As shown in the above plot, there does not appear to be a statistically-significant divergence in the coefficients of treated and control units during the pre-treatment period from 1987-1998. This suggests that the parallel trends assumption is not unreasonable in this scenario.

Part F

Here, we use `feols` to apply the TWFE estimator to our original dataset. This dataset contains a much larger sample of towns which can receive treatment at different points in time.

```
# Fit TWFE model

mod_1F <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = data,
  vcov = cluster ~ swis_code
)

# Extract ATT estimate and SE
ATT_1F <- mod_1F$coefficients["appointed"]
SE_1F <- mod_1F$se["appointed"]

# print results to console
summary(mod_1F)
```

```

OLS estimation, Dep. Var.: reassessed
Observations: 12,750
Fixed-effects: swis_code: 510, year: 25
Standard-errors: Clustered (swis_code)
              Estimate Std. Error t value Pr(>|t|)
appointed    8.61205    1.83572 4.69137 3.4918e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 33.8      Adj. R2: 0.266628
              Within R2: 0.005053

```

```
glue("ATT estimate produced TWFE model and original dataset: {round(ATT_1F,6)}")
```

ATT estimate produced TWFE model and original dataset: 8.612045

```
glue("Standard error: {round(SE_1F,6)}")
```

Standard error: 1.83572

This approach is likely to result in biased estimates of the ATT because the TWFE estimator cannot accommodate staggered adoption of treatment; this occurs due to “forbidden comparisons” or “negative weights” that implicitly use already-treated units as controls for newly-treated units.

Part G

Here, we use `feols` to fit the following outcome model using only untreated observations:

$$Y_{it}(0) = \alpha_i + \gamma_t + \epsilon_{it} \forall (i, t) \notin \mathcal{M}$$

Where $\mathcal{M}$ is the set of treated observations. The fitted model is then used to impute the counterfactual outcome and individualistic treatment effect for treated observations:

$$\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$$

The ATT is subsequently estimated by taking an average of the individualistic treatment effects for treated observations:

$$\hat{ATT} = \frac{1}{|\mathcal{M}|} \sum_{(i,t) \in \mathcal{M}} \hat{\tau}_{it}$$

```
# Split data into set of treated / untreated observations
untreated_obs <- data[data$appointed == 0,]
treated_obs <- data[data$appointed == 1,]

# Fit TWFE model for untreated potential outcome
mod_1G <- feols(
  fml = reassessed ~ 1 | swis_code + year,
  data = untreated_obs,
  vcov = cluster ~ swis_code
)
```

NOTE: 5/0 fixed-effect singletons were removed (5 observations).

```
# Use fitted model to impute counterfactual outcome for treated observations
treated_obs$reassessed_ct <- predict(mod_1G, newdata = treated_obs)

# Compute individualistic treatment effect (ITE)
treated_obs$ITE <- treated_obs$reassessed - treated_obs$reassessed_ct

# Compute ATT as average of ITEs
ATT_1G <- mean(treated_obs$ITE, na.rm=TRUE)

glue("ATT estimate produced via manual implementation of FEct estimator: {round(ATT_1G,6)}")
```

ATT estimate produced via manual implementation of FEct estimator: 9.89246

Part H

Here, we use the `fect` package to estimate the following quantities:

1. The “average” ATT over the entire study period
2. The “dynamic” ATT stratified by the number of years since treatment adoption

```
# set seed for reproducibility
set.seed(42) # fect uses bootstrapping for variance estimation

# fit fixed-effects counterfactual estimator model
mod_1H <- fect(
  formula = reassessed ~ appointed,
  data = data,
```

```

index = c("swis_code", "year"),
method = "fe",
force = "two-way",
se = TRUE,
nboots = 1000
)

```

```

+-----+
| Parallel computing: using 1 of 2 available cores. |
|                                                    |
| To change: set cores = <n> in fect().             |
| Default: min(available - 2, 8).                   |
+-----+

```

Loading required package: foreach

Attaching package: 'foreach'

The following objects are masked from 'package:purrr':

accumulate, when

```

# print results to console
mod_1H

```

Call:

```

fect.formula(formula = reassessed ~ appointed, data = data, index = c("swis_code",
  "year"), force = "two-way", method = "fe", se = TRUE, nboots = 1000)

```

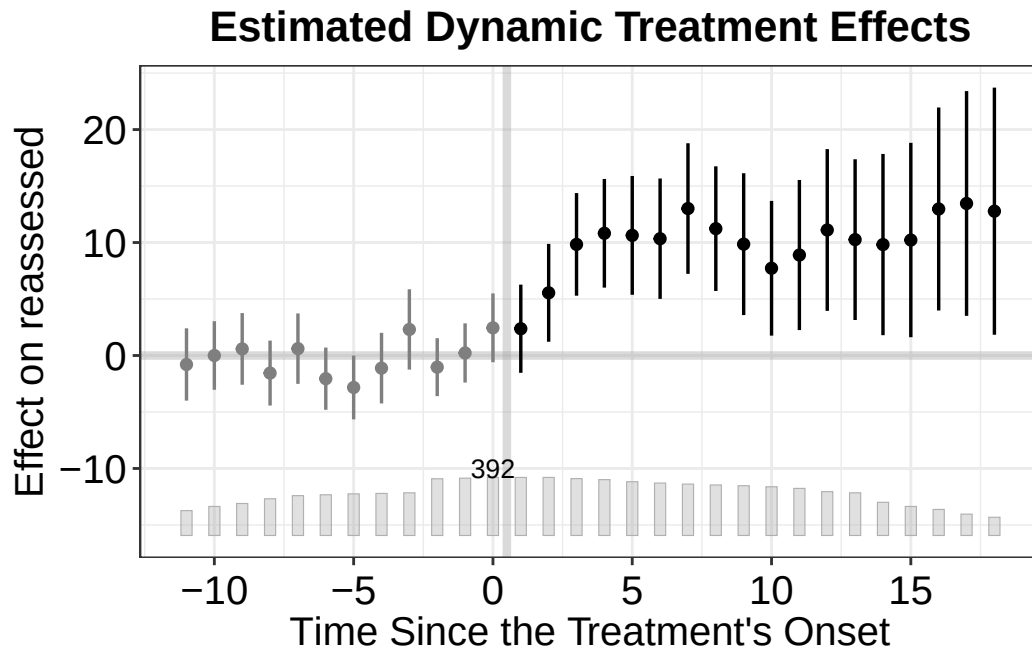
ATT:

	ATT	S.E.	CI.lower	CI.upper	p.value
Tr obs equally weighted	9.722	2.148	5.513	13.93	5.990e-06
Tr units equally weighted	10.761	2.195	6.460	15.06	9.426e-07

```

# plot fect results
plot(mod_1H)

```



The above plot of dynamic treatment effects (which appear to be non-significant during pre-treatment periods) suggests that the estimator's assumption of no pre-trends might be reasonable. However, we can test this assumption in a more robust manner using placebo testing.

Part I

Here, we conduct a placebo test with three placebo periods to check for the presence of pre-trends.

```
# Conduct placebo test
mod_1I <- fect(
  formula = reassessed ~ appointed,
  data = data,
  index = c("swis_code", "year"),
  method = "fe",
  force = "two-way",
  se = TRUE,
  nboots = 1000,
  placeboTest = TRUE,
  placebo.period = c(-2, 0),
)
```

```

+-----+
| Parallel computing: using 1 of 2 available cores. |
| |
| To change: set cores = <n> in fect(). |
| Default: min(available - 2, 8). |
+-----+

```

```

# print results to console
mod_1I

```

Call:

```

fect.formula(formula = reassessed ~ appointed, data = data, index = c("swis_code",
  "year"), force = "two-way", method = "fe", se = TRUE, nboots = 1000,
  placeboTest = TRUE, placebo.period = c(-2, 0))

```

ATT:

	ATT	S.E.	CI.lower	CI.upper	p.value
Tr obs equally weighted	12.13	2.729	6.785	17.48	8.751e-06
Tr units equally weighted	12.77	2.805	7.272	18.27	5.297e-06

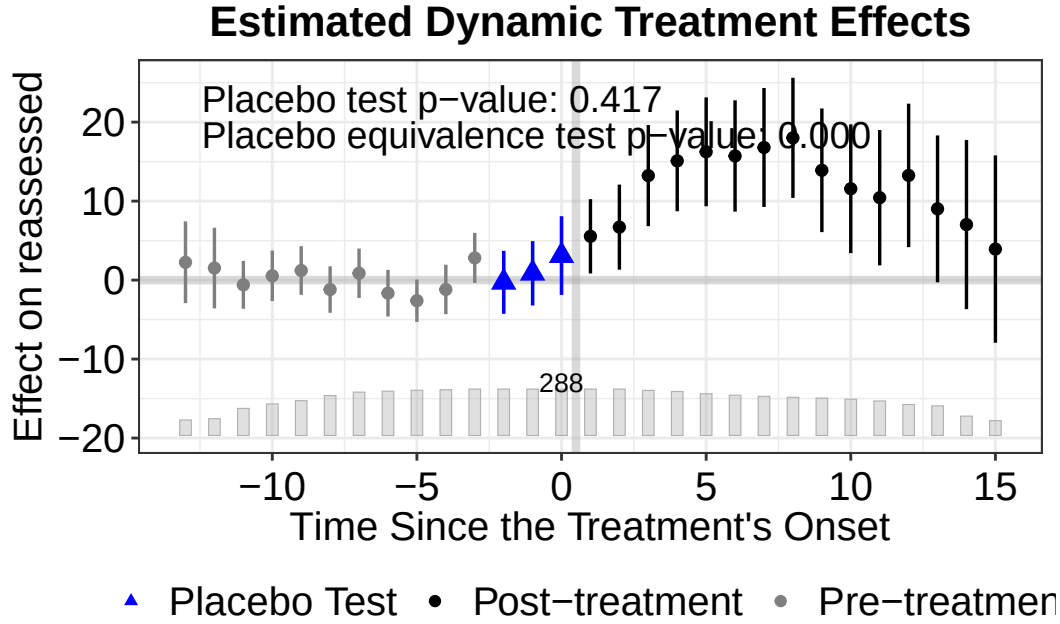
Placebo effect for pre-treatment periods:

	ATT.placebo	S.E.	CI.lower	CI.upper	p.value	CI.lower(90%)	CI.upper(90%)
[1,]	1.228	1.511	-1.734	4.19	0.4165	-1.258	3.713

```

# plot fect results
plot(mod_1I)

```



The built-in placebo test procedures of `fect` return two main diagnostic measures:

1. The placebo test p-value. This p-value represents the probability of observing our effect size under the null hypothesis of $ATT = 0$ during pre-treatment placebo periods (i.e., no pre-trends). If this p-value is significant, then it indicates that the assumption of no pre-trends is violated.
2. The placebo equivalence test p-value or “two one-sided tests” (TOST) p-value. This p-value represents the probability of observing near-zero effects during all pre-treatment placebo periods under the null hypothesis that the $ATT < -\theta_2$ or $ATT > \theta_1$ where $[-\theta_2, \theta_1]$ represents the range of values that the researcher deems to be sufficiently close to zero (i.e., the equivalence range). By default, `fect` sets this range as $[-0.36\hat{\sigma}_\epsilon, 0.36\hat{\sigma}_\epsilon]$, where $\hat{\sigma}_\epsilon$ is the estimated standard deviation of the residualized untreated outcome, but users can modify this parameter using the `tost.threshold` argument. If this p-value is significant, then it indicates that any pre-trends that might be present are small enough that they fall within the researcher’s tolerance.

Our placebo test procedure found that p-value #1 was insignificant and p-value #2 was significant, both of which support the assumption of no pre-trends.